

What we want to build

We want to make a small first-person escape-room game set in an Ancient Egyptian tomb. The player is an archaeologist who steps into a burial chamber and accidentally triggers the closing mechanism. The whole game is about getting back out.

We're aiming for around 5–8 minutes of gameplay across 2–3 small, connected chambers.

Multimedia types and how they show up in the game

Type	What it is in the game	How the player interacts with it
Image	Images of hieroglyphs	Interaction with images to solve a puzzle
Video	Short cut-scene	Plays once in the beginning
Audio	Ambient (torch crackle, footsteps), music, audio input	Volume slider in the UI, separate slider for music and sfx, microphone
Text	Text that comments on what the player should/did do	Text changes based on which puzzle is active/solved
Niagara	Sand, portals	Effects appear or change in response to puzzle state

One feature per category (the six we picked)

Category	Feature
World builder tweaks	Skybox and baked lights with Lumen (torches, sky) Landscape tool (before entering the tomb)
Text/UI	Text Render Actor (see above) Text Animation (menu)
Audio	Spatial audio (footsteps, torches) Audio Input (to solve puzzles)
Image/UI	Interaction with images updates them to solve puzzles
Video	Cut-scene (in the beginning)
Effects & Programming	Improve character motion feature (to solve puzzle)

Main Menu Level

The main menu serves as the entry point of the application. It presents a stylized background image with three buttons: Play, Settings, and Quit. Play transitions the player to the first puzzle level. Settings opens a dedicated settings page with volume sliders for Master, SFX, and Ambient audio channels, allowing the player to adjust the sound balance. Quit exits the application. All volume settings are stored in a Game Instance Blueprint (GI_TombEvader), ensuring they persist across level transitions without being reset. Ambient background music plays while the menu is displayed.

Features used in this level:

- Text/UI – Text Animation: All menu buttons feature a hover animation with a subtle upscale and colour change, providing visual feedback to the player.
- Audio: Ambient background music plays in the menu. Volume settings (Master, SFX, Ambient) are adjustable through sliders and applied globally via Sound Mix Class Overrides.
- Custom Blueprints: WBP_MainMenu, WBP_Settings, WBP_PauseMenu, GI_TombEvader. Menu background image is a custom asset.

Desert Puzzle Level

The level begins with an introductory cutscene before the player takes control of a first-person MetaHuman character. The player spawns on a custom desert landscape in front of a pyramid. Upon entering the pyramid, the door behind the player closes with a sound effect. To proceed to the next level, the player must solve a picture based puzzle by interacting with different images and arranging them into a predefined order on the wall. Correctly placed images light up to provide visual feedback. Once the puzzle is completed, a portal appears, allowing the player to travel to the next level.

Features used in this level:

- Video: An introductory cut-scene plays at the start using a CineCameraActor and Level Sequence
- Image/UI: The player interacts with framed hieroglyph images mounted on the wall. Pressing on a picture frame cycles through available images. Placing all images in the correct predefined order causes them to glow, indicating the puzzle is solved.
- Effects & Programming – A sprint button provides faster horizontal movement.
- Audio: Spatial audio with reverb is applied to torch fire and ambient sounds. A closing door sound effect plays when entering the pyramid. Footstep sounds change dynamically based on the surface material (sand outside, stone inside). Portal sound is played when the puzzle is solved (sounds used are free and were downloaded from the internet)
- Niagara: Torch fire particle effects provide ambient lighting. A portal Niagara effect appears when the picture puzzle is solved.
- Landscape: The desert environment outside the pyramid was created using the Landscape tool with sculpted sand dunes.
- Custom Blueprints:
 - BP_Portal – Used stone meshes from fabstore to build the frame. Everything else is custom made.
 - BP_Torch – Used mesh from fabstore.
 - BP_StandingTorch – Used mesh from fabstore.
 - BP_Hieroglyph – Completely custom made except of material used for StoneTablet. GIMP was used to create the Hieroglyph
 - BP_PictureFrameInteractable – Completely custom made except of golden material used for the frame. Pictures were generated with ChatGPT and later cropped in GIMP.
 - BP_PictureManager – only holds puzzle logic and is completely custom made.
 - BP_TombEntrance – The doors mesh is from fabstore, all the rest is implemented by us.

Void Puzzle Level

The player starts in a dark room with a glowing hint text displayed as a Text Render Actor. The text reacts to the player's microphone input by glowing brighter when the player speaks. The core mechanic of this level is voice-based interaction: rune stones are placed around the room, and the player must speak near them to activate them. An Audio Listener Blueprint captures microphone input and triggers the corresponding lights to light up. The player must escape the labyrinth by finding all lights to guide the correct way. In the end a Portal appears to go to the last level.

Features used in this level:

- Audio Input / Microphone: The player uses voice to interact with lights. Microphone input is captured and lights are activated based on proximity.
- Text/UI – Text Render Actor: A reactive Text Render Actor at the start of the level provides a hint and glows in response to the player's voice input, reacting to what the player is doing.
- Labyrinth – was created with meshes from the fabstore. (Created a cube with walls, copied all walls and placed them in top-down view. Later deleted walls to create corridors)
- Effects & Programming – A sprint button provides faster horizontal movement.
- Custom Blueprints:
 - BP_AudioListener – initially a manager blueprint, now creates the widget for microphone level
 - BP_GoalBox – box collider that makes the portal appear
 - BP_InteractiveTextRender – Except Font, completely custom made. (Material emissive color based on microphone input)
 - BP_VoiceRune – copied base material from fabstore asset. Enhanced material to also react to microphone input.

Sky Puzzle Level

The final level takes place in an open sky environment with a completely different atmosphere from the previous levels. The player must navigate a series of different platforms to reach the goal and complete the game. An Ascension pedestal is placed in the beginning of the level to provide a jump boost when interacting with it, letting the player jump higher and navigate more freely in air to get to the next platform. A sprint button is available to help with the platforming sections. If the player falls, a respawn volume catches them and teleports them back to the beginning.

Features used in this level:

- World Builder – Skybox & Lights: A custom skybox with modified light parameters
- Landscape: A custom terrain was created using the Landscape tool for the base environment.
- Effects & Programming – Improved Character Motion: The jump boost mechanic on ascension pedestal (BP_AscensionPadestal) enhances the player's vertical movement. A sprint button provides faster horizontal movement for platforming.
- Audio: Spatial audio and surface-based footstep sounds continue from previous levels.
- Custom Blueprints:
 - BP_AscensionPadestal – Constructed using meshes from fabstore.
 - BP_RespawnVolume – Collider used for respawning the player when falling of an platform.
- Custom CloudPlane – This plane uses a custom material that uses noise to create custom clouds reaching to the horizon from which the Skybox takes over so it looks like you're high up in the sky.

General assets from Fab

Stylized Egypt pack, Medieval Torch, Essential_Foosteps_SK, Marble_Altar, TorchFire, Anubis, Guardians_of__Stone_, Egyptian Papyrus